

Generative Scientific Discovery Engine

Technical Report

Namit Aggarwal

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Abstract

Given 400 sparse, noisy observations of a field $u(x, t)$ and no governing equation, the system retrieves scientific evidence (RAG + knowledge graph), generates candidate equations (LLM + graph templates), tests each with an inverse physics-informed neural network (PINN), rejects invalid hypotheses on held-out data, selects the most parsimonious supported law, and predicts beyond the observed window. It recovers $u_t = D u_{xx}$ with $\hat{D} = 0.10002$ (truth 0.1), rejects advection and pure reaction, and predicts unseen times with RMSE 2.5×10^{-4} , $40\times$ better than a physics-free network. On a second, unseen system it recovers $u_t + c u_x = D u_{xx}$ with both constants within 0.4%. It passes 50/50 benchmark questions, 23/25 multi-hop problems and 20/20 adversarial cases, and the pipeline also runs end to end with a real LLM (Google Gemini).

1 System overview

The pipeline follows the brief: *Observed data* $\rightarrow$ *evidence* $\rightarrow$ *candidate physics* $\rightarrow$ *equation* $\rightarrow$ *PINN* $\rightarrow$ *validation* $\rightarrow$ *new prediction* (Figure 1). Division of responsibility is enforced in code: RAG retrieves, the graph organises, the LLM only *proposes*. Accept/reject and selection use only measured losses on held-out noisy observations; the hidden truth is read in one function, after all decisions, to report error.

Data. Ground truth $u_t = 0.1 u_{xx}$ on $x, t \in [0, 1]$, zero boundary values, $u(x, 0) = \sin \pi x + 0.5 \sin 3\pi x$. The system sees only 400 random points with 5% Gaussian noise. Splits: train on $t \leq 0.8$ (minus a random 20% holdout used for the gate and selection); OOD = all $t > 0.8$, never used for any decision; novel prediction on $t \in (1, 1.5]$, where no data exists. All thresholds were fixed in `config.yaml` before each run.

2 Requirement coverage

Minimum research challenge (all met).

Required	Delivered
≥ 3 generated hypotheses	5 candidates (mock run), 6 (Gemini run)
1 discovered governing equation	$u_t = D u_{xx}$, $\hat{D} = 0.10002$; and $u_t + 0.399 u_x = 0.0201 u_{xx}$ on a second system
1 inverse / 1 forward PINN	Phase 3 (both), every Phase 6 candidate is an inverse PINN
1 sparse/noisy experiment	all main experiments use 400 points with 5% noise
1 OOD prediction	$t > 0.8$: RMSE 0.0018 vs clean field; novel $t \in (1, 1.5]$: 2.5×10^{-4}
1 correctly rejected hypothesis	advection (holdout RMSE 0.227 vs gate 0.03) and pure reaction (0.118)
20 adversarial physics cases	20/20 (3 by abstention)

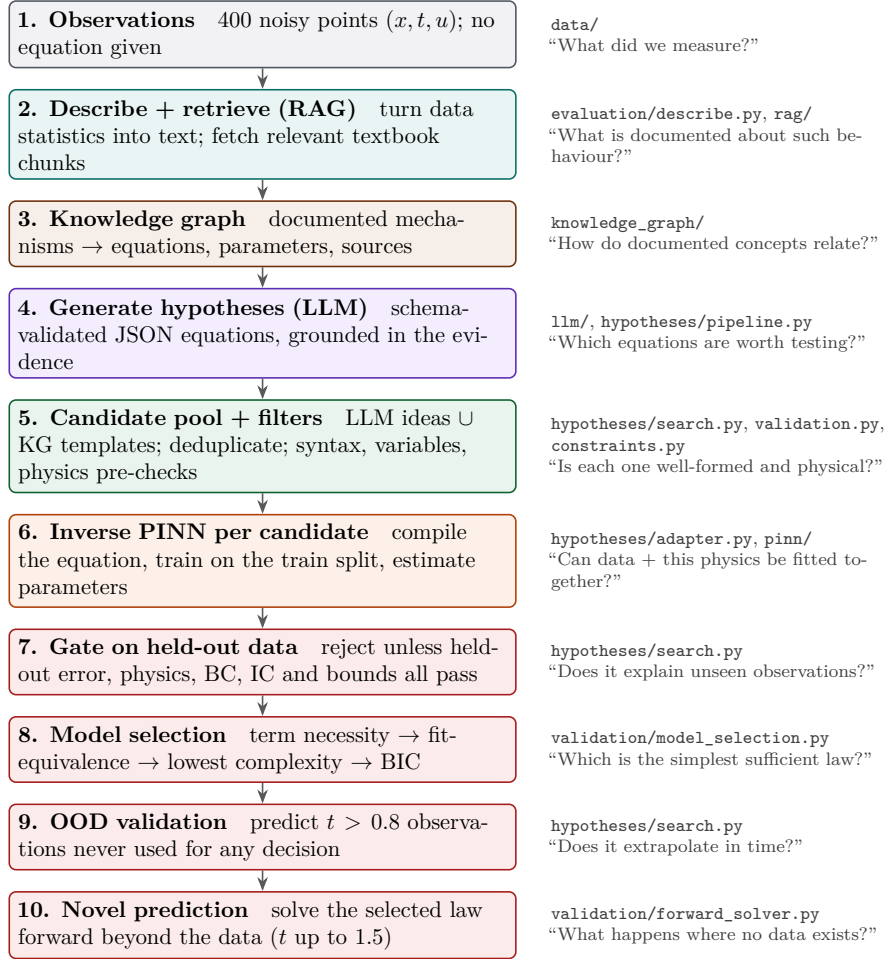


Figure 1: End-to-end pipeline. Colours mark the code package that implements each stage.

Mandatory research components

Mandatory component	Implementation and evidence	Status
Scientific RAG + knowledge graph	<code>rag/</code> , <code>knowledge_graph/</code> ; 22 entities, 40 relationships, all with provenance	Met
Generative hypotheses from multi-source evidence	LLM proposals $\cup$ graph templates (<code>hypotheses/search.py</code>); mock and real Gemini runs	Met
PINN forward and inverse	<code>pinm/</code> ; forward RMSE 7.3×10^{-4} ; inverse $\hat{D}$ error 0.11%	Met
PDE/ODE discovery from data	SINDy (<code>equation_discovery/</code>) on clean/sparse data; PINN-based identification on noisy data	Partial
Unknown parameter estimation (PINN)	D within 0.02%; c, D within 0.4% on a second system	Met
Sparse/noisy-data learning	400 points, 5% noise, two systems	Met
Boundary/initial-condition inference	BC/IC supplied as known physics; enforced and checked, not inferred	Not done
Physics-constrained generative search	bounds ($D, K > 0$), well-posedness pre-checks, PINN gate (Figures 2-3)	Met
Uncertainty estimation	bootstrap coefficient stability (20 subsets); not Bayesian	Partial
Out-of-distribution validation	held-out $t > 0.8$ and novel window $t \in (1, 1.5]$	Met
Comparison with NN and numerical solver	same network without physics; RK4/Euler forward solver	Met

3 Method: physics-constrained generative search

1. **Evidence.** The observations are summarised as text (statistics only, never naming a mechanism) and used to query a 5-document corpus; the knowledge graph returns the documented equations and parameters. A check confirms the corpus never states the dataset’s parameter values (no answer leakage).
2. **Generation.** The LLM returns schema-validated JSON hypotheses; graph templates are added; structural duplicates are removed (parameter- and sign-invariant signatures).
3. **Pre-checks.** Syntax, variables, parameters, declared vs. actual derivatives, unit metadata, evolution form, physical bounds, boundary-condition well-posedness.
4. **Inverse PINN per candidate.** Each equation string is compiled by SymPy into the residual of one shared PINN (4×64 tanh, Adam 3000 + L-BFGS 300); unknown parameters train jointly with the network.
5. **Gate** (held-out noisy data): holdout RMSE ≤ 0.03 , physics residual ≤ 0.01 , BC loss ≤ 0.005 , IC loss ≤ 0.01 , physical bounds.
6. **Selection** (Figure 3): fit-equivalence (within 5% of best) $\rightarrow$ drop inactive terms ($<5\%$ of u_t) $\rightarrow$ lowest complexity $\rightarrow$ BIC.
7. **Validation and prediction.** Bootstrap stability, conservation check, OOD prediction, and a forward solve of the selected law beyond the data.

Ranking criteria requested by the brief, and where each is used:

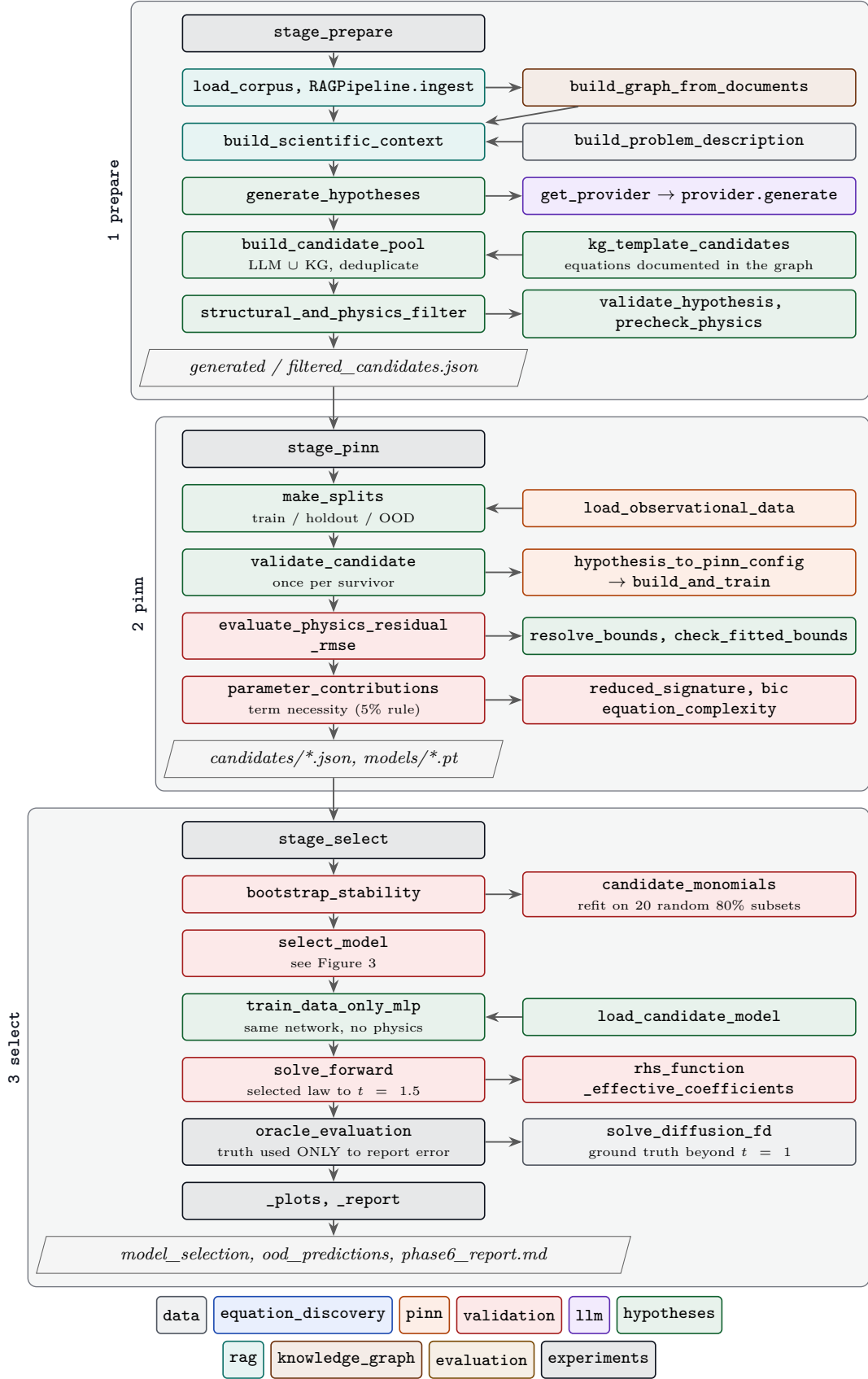


Figure 2: Function-level flow of the search (experiments/08_physics_constrained_search.py), stage by stage. Only oracle_evaluation reads the ground truth, after all decisions.

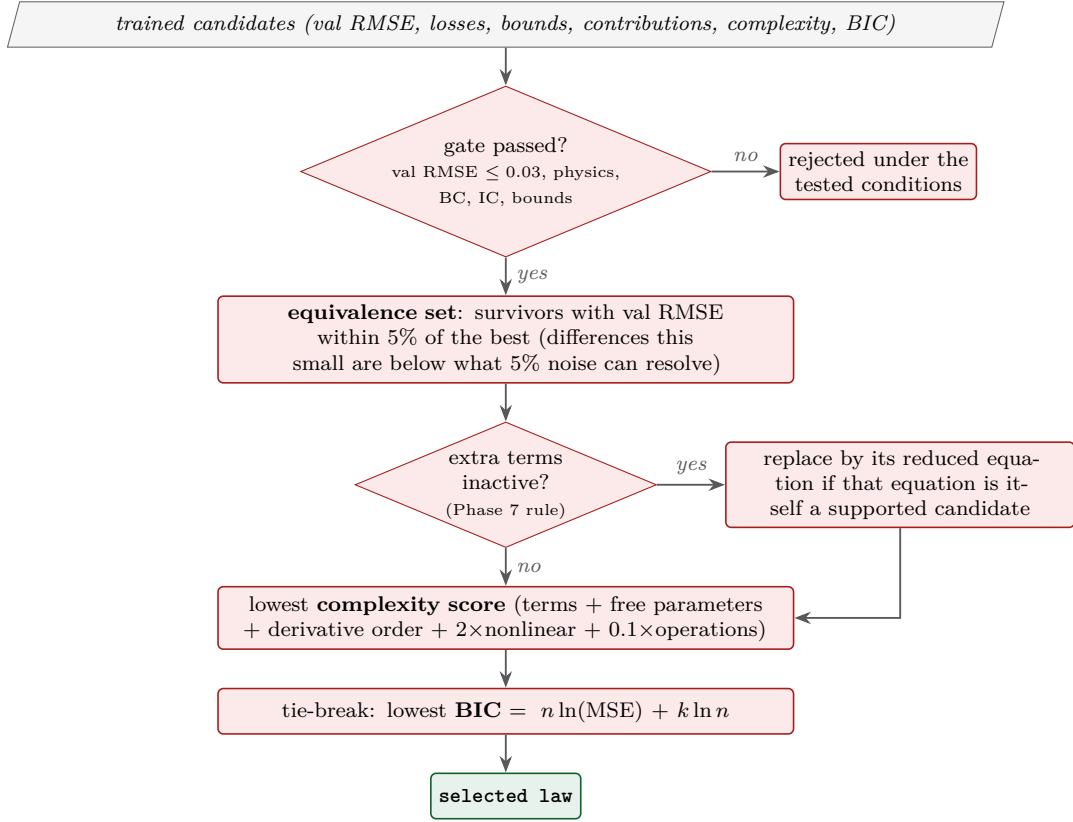


Figure 3: Model-selection rule (`validation/model_selection.py`); thresholds fixed before any run.

Criterion	Measure	Used in
Data fit	holdout RMSE on noisy observations	gate, equivalence
PDE residual	mean squared residual on a fresh grid	gate
Conservation laws	global balance violation $\text{RMS}_t \int \mathcal{R} dx / \text{RMS}_t dM/dt$	reported
Dimensional consistency	declared units per parameter (fraction complete)	pre-check (metadata)
Boundary/initial conditions	BC and IC losses; well-posedness warnings	gate, pre-check
Complexity	terms + parameters + derivative order + 2×nonlinear + 0.1×ops	selection
Uncertainty	bootstrap coefficient mean, spread, sign stability	reported

4 Results

4.1 Hypothesis testing and the discovered law

ID	Candidate	Estimates	Holdout	Status	Inactive	Cons. viol.
H1	$u_t = D u_{xx}$	$D = 0.10002$	0.01214	selected	–	0.08%
H2	$u_t + c u_x = 0$	$c = 0.0089$	0.2269	rejected	–	100%
H3	$u_t = D u_{xx} + r u(1 - u/K)$	$D=0.101, r=0.023, K=1.18$	0.01217	supported	r, K	0.11%
H4	$u_t + c u_x = D u_{xx}$	$c=-0.0002, D=0.09995$	0.01214	supported	c	0.13%
K3	$u_t = r u(1 - u/K)$ (graph)	$r=-1.44, K=6.67$	0.1176	rejected	–	0.75%

H1, H3 and H4 fit to the noise floor (≈ 0.012); H3 and H4 carry inactive terms and reduce to H1, which has the lowest complexity, and BIC agrees. A pure best-fit rule would have chosen H4 (lower by 0.06%). Advection breaks global conservation completely: with zero boundary values it conserves $\int u dx$, while the data loses it.

4.2 PINN research results

Metric	Forward (D fixed)	Inverse (D from 0.2)	Phase 6 selected (H1)
PDE residual (fresh grid)	2.7×10^{-5}	3.3×10^{-5}	1.4×10^{-5}
Data loss	1.32×10^{-4}	1.32×10^{-4}	1.29×10^{-4}
Boundary / initial loss	$2.4 \times 10^{-6} / 8 \times 10^{-7}$	$2.8 \times 10^{-6} / 1.2 \times 10^{-6}$	$1.0 \times 10^{-6} / 1.0 \times 10^{-6}$
Parameter error	–	0.11%	0.02%
Prediction error (vs clean)	RMSE 7.3×10^{-4}	RMSE 8.8×10^{-4}	OOD RMSE 1.8×10^{-3}
Conservation violation	–	–	0.08% (train), 1.7% (OOD), 5.6% ($t > 1$)
Uncertainty (bootstrap u_{xx} coef.)	–	–	0.110 ± 0.003 , sign-stable 100%
OOD (noisy holdout, $t > 0.8$)	–	–	RMSE 0.0114 (noise floor ≈ 0.012)

The bootstrap uses finite-difference derivatives and is biased high (0.110 vs. the PINN’s 0.100); it measures term stability, not a calibrated interval.

4.3 Comparison with a conventional network and a numerical solver

Model	OOD RMSE, $t > 0.8$ (vs clean)	Conservation viol. (train / OOD)
PINN, selected law H1	0.0018	0.08% / 1.7%
Same network, data only (no physics)	0.0712 (40× worse)	15% / 5.3%
Forward solver (RK4) with $\hat{D}$, no data	0.0003; novel $t \in (1, 1.5]$: 0.00025	–

4.4 Sparse/noisy data and generalisation

SINDy recovers $u_t = 0.1009 u_{xx}$ (clean) and $0.1013 u_{xx}$ (sparse) but fails on the 400 noisy points (five spurious terms); the PINN path succeeds on the same data. On a second hidden system ($u_t + 0.4 u_x = 0.02 u_{xx}$, $x \in [0, 1.5]$, Gaussian start, 400 points, 5% noise) the identical pipeline rejects diffusion, advection and reaction–diffusion and selects advection–diffusion with $c = 0.39897$, $D = 0.02008$ (errors 0.26%, 0.40%), OOD RMSE 0.0010 vs. 0.0068 for the physics-free network, and 0.58% error beyond the data window. SINDy on that data returned a spurious equation.

4.5 Real LLM (Gemini)

With `gemini-3.5-flash-lite` (larger models were overloaded), retrieved evidence changed the generated hypotheses on all three test datasets. In the full search Gemini proposed Du_{xx} , $Du_{xx} - ru$, advection–diffusion and reaction–diffusion; the graph added pure advection and pure reaction, which were rejected. The same law was selected ($\hat{D} = 0.10001$, OOD RMSE 0.0020 vs. clean). This is a single sample of a stochastic generator.

5 Benchmark and ablation

Twelve benchmark datasets (diffusion, advection, reaction–diffusion, advection–diffusion; 3 settings each). **50 questions: 50/50; 25 multi-hop problems: 23/25** (both failures from top-1 lexical retrieval); **20 adversarial cases: 20/20**, including invalid equations, negative diffusivity, time-reversed data, disguised duplicates, malicious LLM output, a non-identifiable dataset, Occam traps, answer leakage and prompt injection (three noise/sparsity cases pass by abstention).

Ablation arm (12 datasets)	Correct law	OOD within 5%
LLM/RAG generation, no validation	3/12	3/12
RAG + KG generation, no validation	3/12	3/12
Equation discovery alone (SINDy)	8/12	6/12
Full system (generation + physics validation + selection)	12/12	12/12
Full system without the LLM	12/12	12/12

Physics validation and selection drive accuracy; RAG supplies a candidate pool containing the truth (top-1 retrieval only 1/6); the graph and mock LLM add no measurable accuracy here. A leakage audit confirms no decision function reads ground truth and catches a planted cheating selector.

6 Limitations

- Synthetic 1-D PDEs only; known laws recovered, no new law claimed.
- Boundary/initial conditions are given, not inferred.
- Uncertainty is a bootstrap of term stability, not a Bayesian posterior.
- Nested models fit equally well; selection relies on a stated parsimony rule, not uniqueness.
- Lexical retrieval over a 5-document corpus; thresholds hand-set (fixed in advance, not calibrated).
- Conservation is a consistency check, not a discriminator: a network fitted to a wrong law can still follow that law (K3: 0.75%), so it is reported alongside the data gate rather than replacing it.

7 Deliverables and reproducibility

Artifact	Location
Code	data/, equation_discovery/, pinn/, validation/, llm/, hypotheses/, rag/, knowledge_graph/, evaluation/; 95 tests in tests/
Datasets	data/generated/*.npz (clean, sparse, noisy, benchmark, second system)
Trained models	results/phase6/models/, results/evaluation/generalization/models/
Discovered equations	results/phase6/model_selection.json, .../generalization/generalization_results.json
Knowledge graph	results/rag_kg/knowledge_graph.json
Experiments	experiments/02–12 (one script per phase)
Benchmark	results/evaluation/ (questions, multi-hop, adversarial, ablation, leakage)
Conservation; Gemini run	results/evaluation/conservation/; results/evaluation/real_llm_gemini/
Reports	this report; results/final_report.md; docs/project_guide.pdf (detailed guide)

Reproduce: install with `pip install -r requirements.txt` and test with `python -m pytest tests`. Then run `data/generate_diffusion.py` and the experiment scripts 02–12 in order, each with `--config config.yaml` (each PINN takes about 4 minutes on CPU; stages are cached). The Gemini run needs `GEMINI_API_KEY`; see `README.md`.

A Additional flowcharts

Function-level flows of the remaining components. Colours follow the package legend.

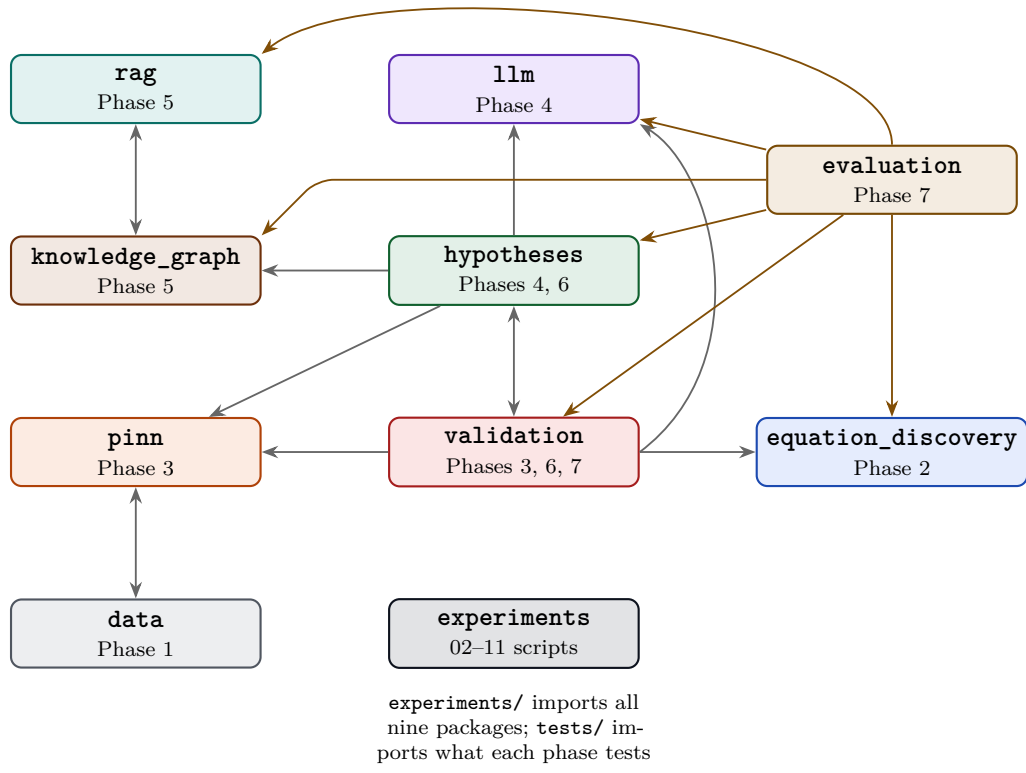


Figure 4: Package dependency map (from the actual imports).

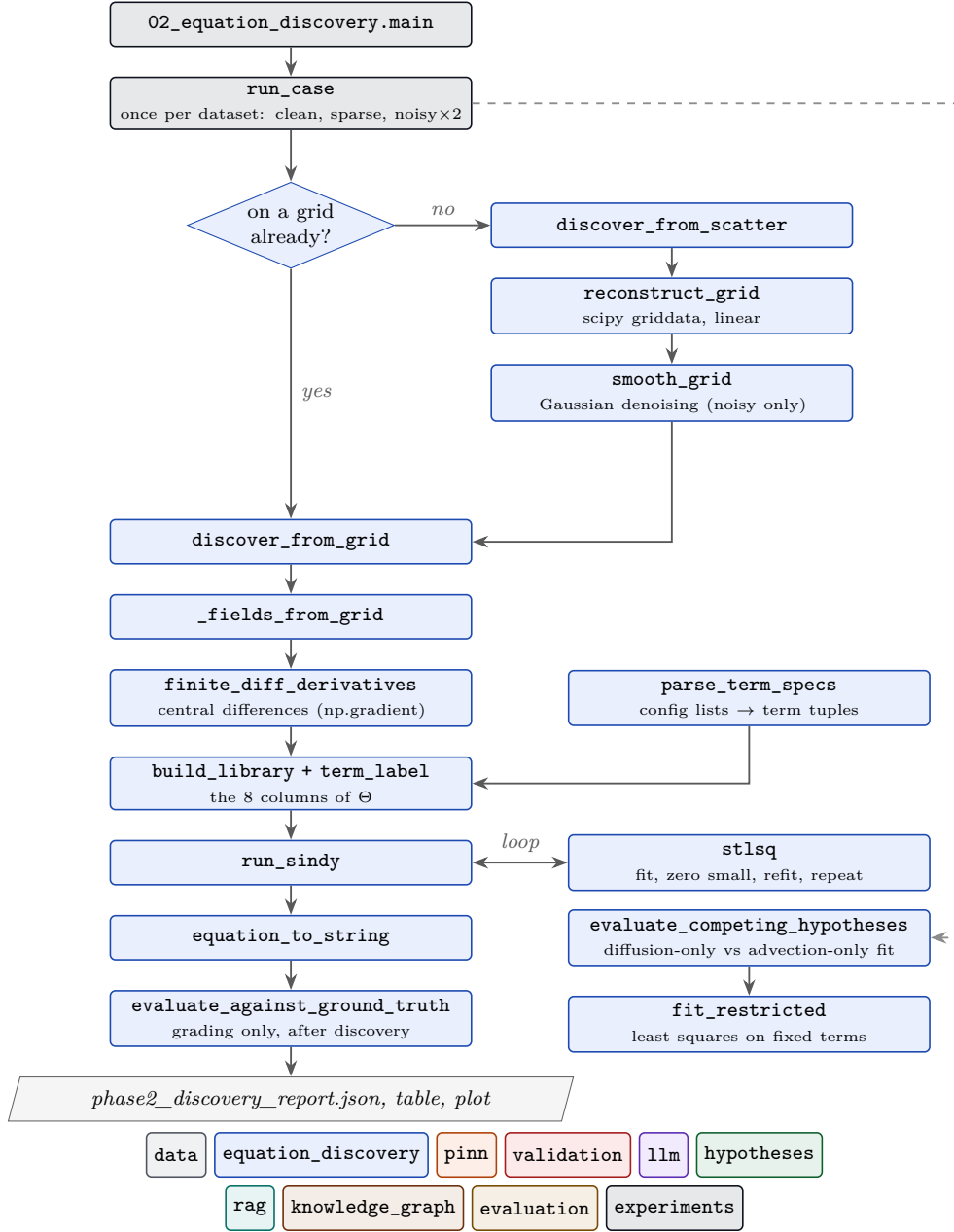


Figure 5: PDE discovery from data (SINDy/STLSQ).

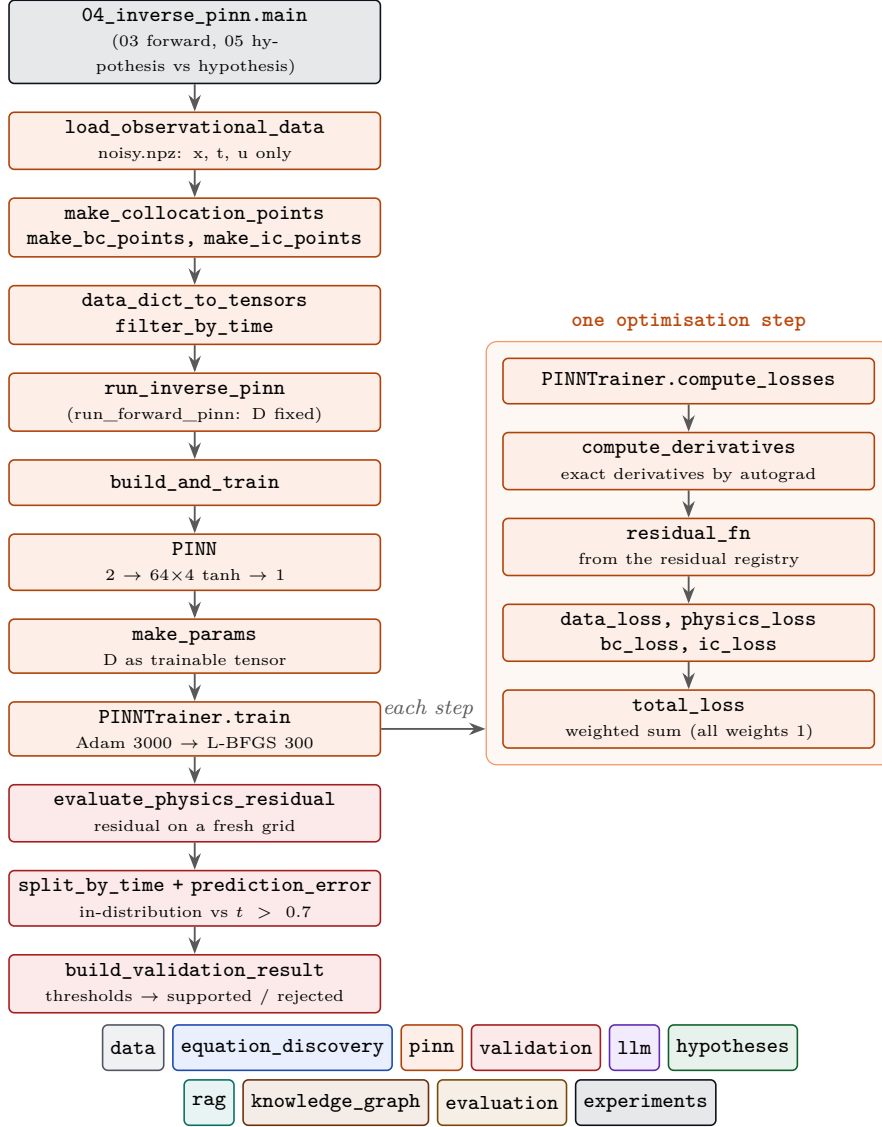


Figure 6: PINN training (forward/inverse) and threshold scoring.

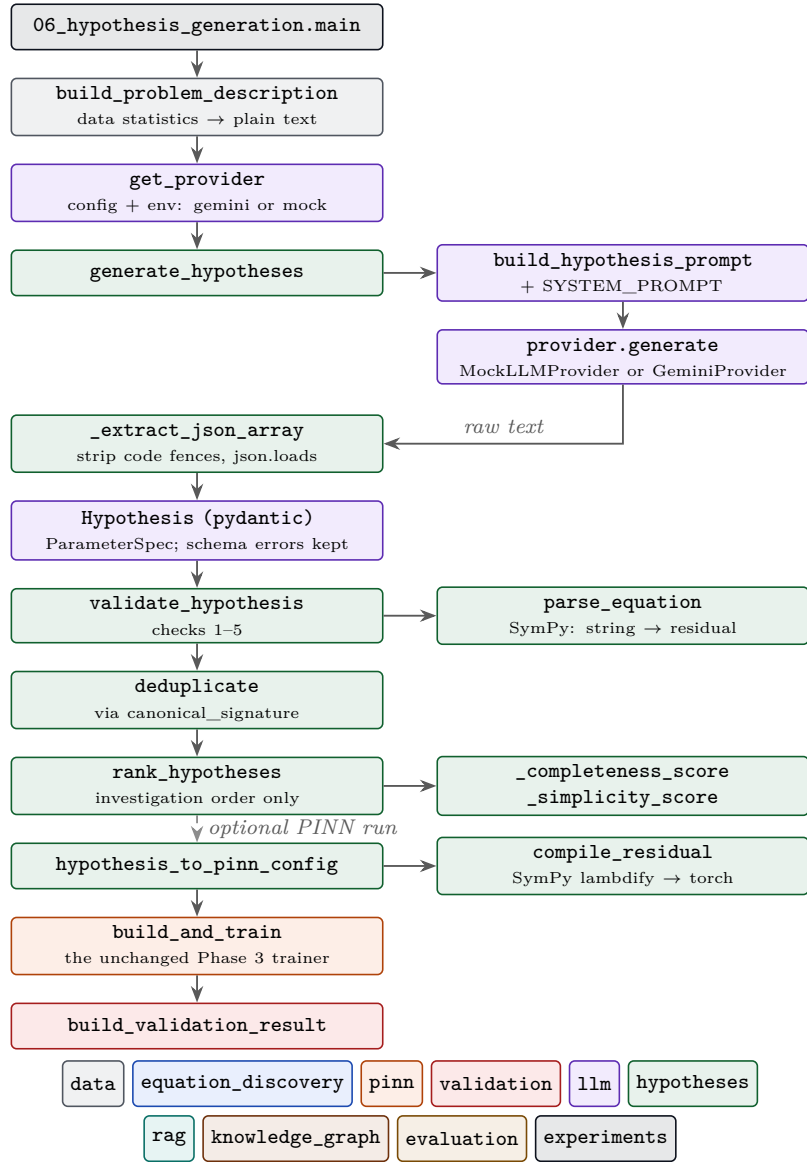


Figure 7: Hypothesis generation, deterministic validation and compilation into a PINN.

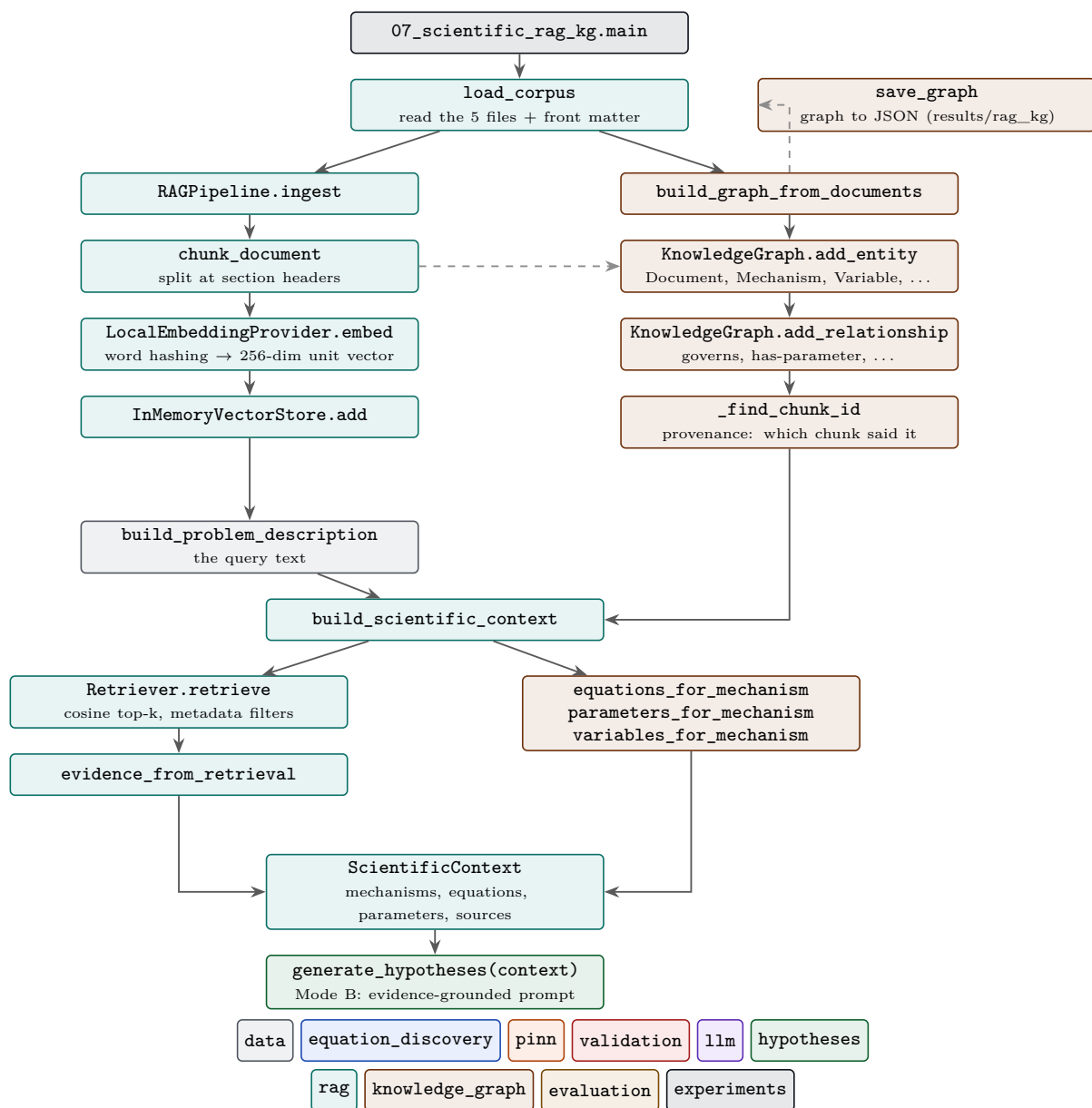


Figure 8: Scientific RAG and knowledge-graph construction into one scientific context.

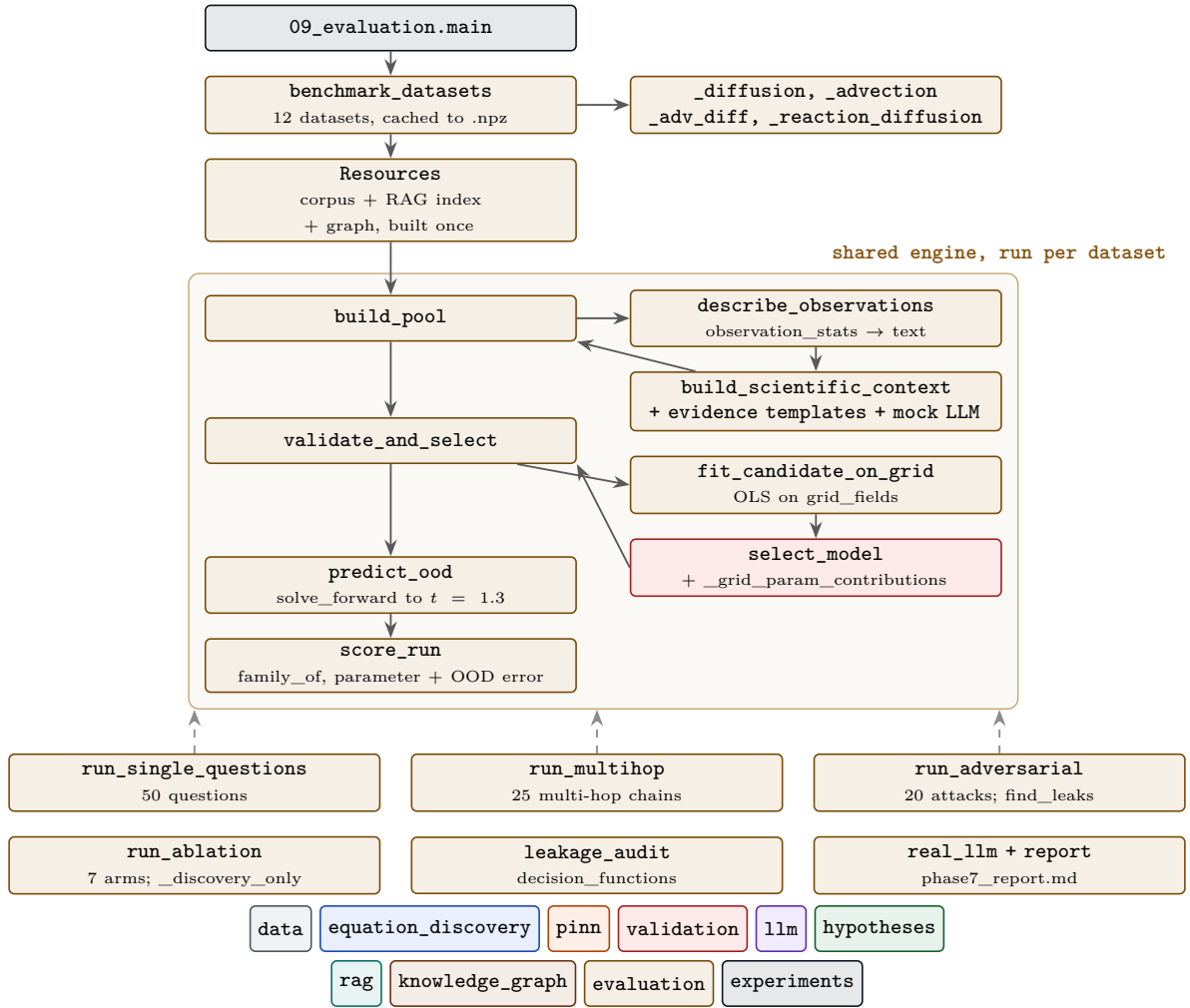


Figure 9: Benchmark, adversarial, ablation and leakage evaluation.

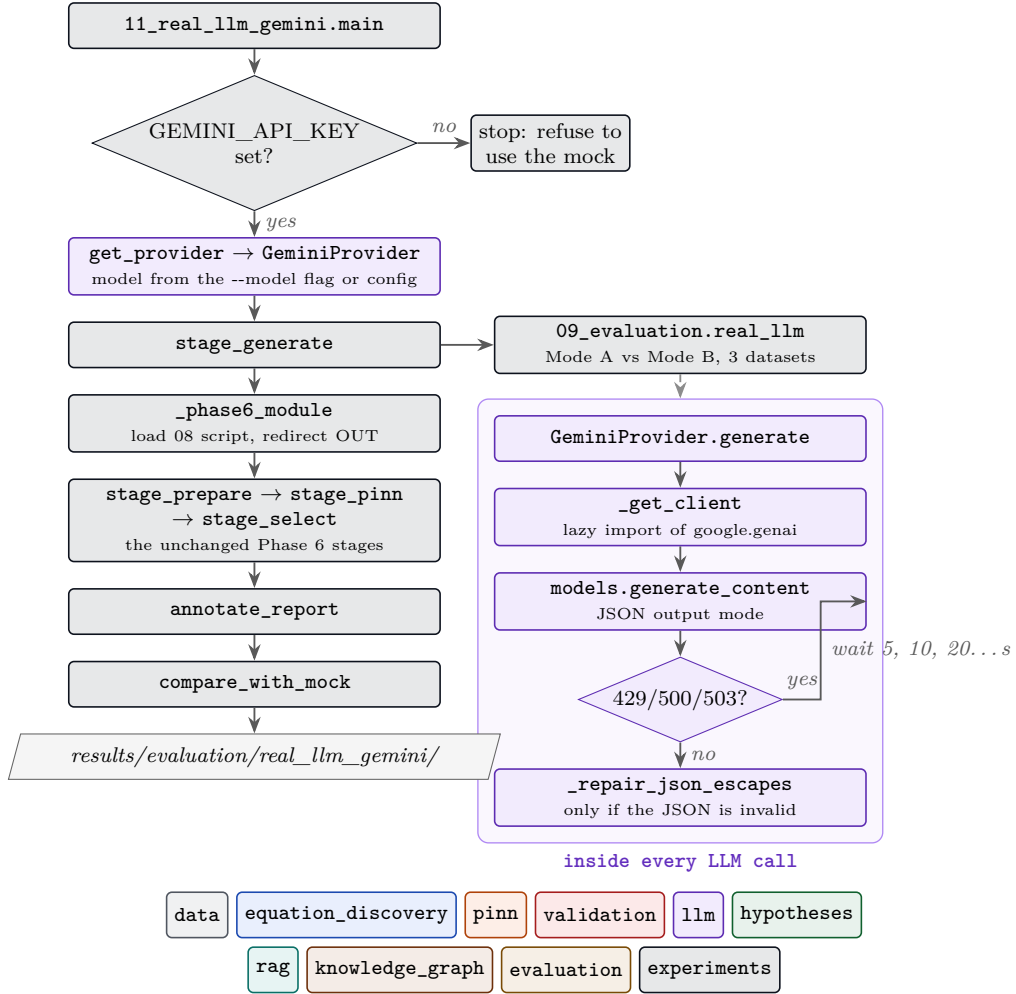


Figure 10: Real-LLM (Gemini) run and the provider’s retry/JSON handling.